

Orthogonal Dual-Tree Topologies for Fault-Tolerant Distributed Aggregation

Dada Khalandhar Gooty

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Abstract

Tree-based communication topologies are widely used in distributed machine learning and distributed aggregation systems due to their logarithmic communication depth and efficient hierarchical structure. However, conventional tree topologies suffer from poor fault tolerance because failure of an internal node disconnects entire dependent subtrees.

This work proposes a preliminary communication topology based on two complementary spanning trees constructed over the same logical node set. The proposed design attempts to reduce persistent node criticality by assigning complementary hierarchical roles across the two trees. Nodes that are highly critical in one tree are intentionally placed in lower-criticality positions in the second tree.

Unlike conventional redundant tree approaches, the proposed topology constructs two structurally distinct overlay trees over the same physical node set. This creates alternate routing structures while preserving hierarchical communication efficiency.

This document presents the preliminary formulation, intuition, structural definitions, and initial theoretical direction of the proposed topology.

1 Introduction

Distributed machine learning systems commonly employ hierarchical communication structures for operations such as:

- gradient aggregation,
- parameter synchronization,
- reduce and all-reduce operations,
- hierarchical broadcast.

Tree-based communication structures are attractive because they provide:

- logarithmic communication depth,
- scalable aggregation,
- low communication overhead.

However, conventional trees suffer from a major limitation:

Failure of an internal node disconnects all dependent descendants.

This creates a significant reliability issue for large-scale distributed systems and distributed machine learning workloads.

This work proposes a preliminary dual-tree topology intended to reduce such failure sensitivity.

2 Dual-Tree Overlay Topology

Let:

$$V = \{v_1, v_2, \dots, v_n\}$$

be the logical node set.

Two tree overlays are constructed over the same node set:

$$T_1 = (V, E_1)$$

and

$$T_2 = (V, E_2)$$

where:

- T_1 and T_2 are complete binary trees,
- both trees span the same physical nodes,
- the edge structures differ in order to diversify dependency relationships.

The combined communication graph is defined as:

$$G = (V, E_1 \cup E_2)$$

Thus each physical node simultaneously participates in both communication trees.

Unlike layered or duplicated graph formulations, nodes are not replicated across trees. The two trees instead represent independent communication overlays constructed over the same distributed system.

The primary objective is to reduce persistent articulation structures by ensuring that nodes occupying highly critical positions in one tree occupy lower-criticality positions in the other.

3 Core Design Principle

The central idea of the proposed topology is:

Node criticality should be anti-correlated across the two trees.

Informally:

- Nodes highly critical in T_1 should become minimally critical in T_2 .
- Nodes near the periphery of T_1 should become structurally important in T_2 .

This attempts to avoid persistent bottleneck nodes across the combined communication structure.

3.1 Role Inversion Intuition

An initial motivating intuition is:

$$L(T_1) \approx I(T_2)$$

$$I(T_1) \approx L(T_2)$$

where:

- $L(T)$ denotes the set of leaf nodes,
- $I(T)$ denotes the set of internal nodes.

The intention is that nodes which are structurally vulnerable in one tree become structurally important in the other tree, and vice versa.

Under idealized constructions:

- internal nodes in one tree contribute degree 3,
- leaf participation in the second tree contributes degree 1,
- producing approximate combined degree 4 for most nodes.

This creates a sparse but significantly more resilient communication structure compared to a conventional single binary tree.

4 Node Criticality

Define a node criticality function:

$$C(v, T)$$

which measures the structural importance of node v within tree T .

Possible definitions include:

- subtree size,
- betweenness centrality,
- ancestor dependency count,
- distance from root.

The proposed topology seeks to minimize persistent criticality across both trees.

A preliminary objective function is:

$$\sum_{v \in V} C(v, T_1) C(v, T_2)$$

Minimizing this objective discourages nodes from simultaneously becoming highly critical in both trees.

5 Structural Diversity

Fault tolerance depends not only on redundancy, but also on structural diversity between the trees.

The proposed topology therefore attempts to minimize overlap between:

- ancestor relationships,
- articulation structures,
- routing dependencies,
- communication bottlenecks.

The intention is that local failures in one tree become non-local in the second tree.

6 Fault-Tolerance Intuition

A single binary tree has vertex connectivity:

$$\kappa(T) = 1$$

because every internal node acts as an articulation point.

Failure of an internal node disconnects all dependent descendants.

The proposed topology attempts to reduce this fragility by combining two structurally distinct binary trees over the same node set.

Suppose a node failure disconnects portions of T_1 .

If dependency relationships in T_2 differ sufficiently, disconnected descendants in T_1 may still remain connected through alternate paths in T_2 .

Thus the combined graph:

$$G = (V, E_1 \cup E_2)$$

may remain connected even after failures that would disconnect a conventional tree.

The topology therefore attempts to eliminate persistent articulation points through dependency diversification across the two tree structures.

7 Failure Model

A physical node failure removes the node from both communication trees simultaneously.

Formally, removal of node:

$$v_i \in V$$

removes:

- all incident edges in T_1 ,
- all incident edges in T_2 .

The resilience objective is therefore not simple redundancy, but rather structural diversification of dependency relationships across the two trees.

8 Preliminary Theoretical Direction

The central theoretical question is the vertex connectivity of the combined graph:

$$G = (V, E_1 \cup E_2)$$

where:

- T_1 and T_2 are complete binary trees,
- both trees span the same node set,
- dependency overlap between the trees is minimized.

A single binary tree satisfies:

$$\kappa(T) = 1$$

due to articulation points.

However, the proposed topology may achieve:

$$\kappa(G) \geq 2$$

or potentially larger connectivity under sufficiently diverse tree constructions.

An important observation is that:

- internal nodes in one tree contribute degree 3,
- leaf participation in the second tree contributes degree 1,
- giving many nodes approximate total degree 4.

This suggests the possibility of significantly stronger connectivity than conventional trees while still preserving sparse communication complexity.

A preliminary conjecture is:

If articulation structures and dependency relationships are sufficiently diversified across the two trees, then the combined graph may eliminate persistent articulation points and achieve substantially improved fault tolerance compared to a single tree.

Formal proof and optimal construction methods remain open problems.

9 Illustrative Example

Figure 1 illustrates the conceptual dual-tree overlay structure.

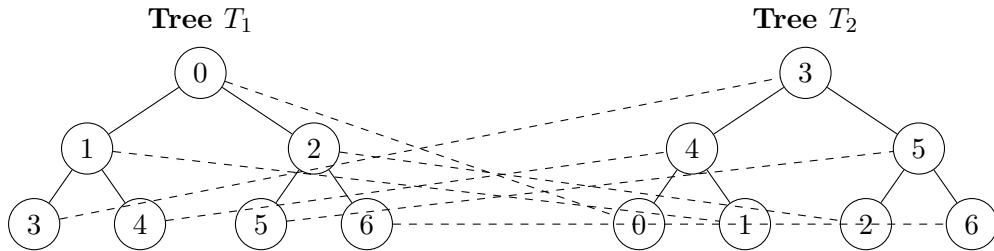


Figure 1: Conceptual dual-tree overlay topology. Both trees are constructed over the same physical node set. Dashed lines indicate correspondence of node identities across the two overlay structures for visualization purposes only.

10 Potential Applications

Potential applications include:

- distributed machine learning,
- hierarchical aggregation,
- fault-tolerant overlays,
- distributed consensus,
- resilient communication fabrics.

11 Open Problems

Several important problems remain open:

- optimal dual-tree construction algorithms,
- formal connectivity proofs,
- minimal-overlap hierarchy generation,
- communication latency analysis,
- dynamic rerouting algorithms,
- comparison with ring and fat-tree topologies,
- resilience under multiple simultaneous failures,
- min-cut and vertex-connectivity characterization,
- articulation overlap minimization.

12 Conclusion

This work introduced a preliminary dual-tree communication topology intended to improve fault tolerance in hierarchical distributed systems.

The key idea is the construction of complementary overlay tree structures with anti-correlated node criticality, reducing persistent bottlenecks and improving alternate communication path availability.

The proposed topology remains an early-stage research direction. Future work includes formal graph-theoretic analysis, construction algorithms, simulations, and distributed machine learning evaluation.

References

- [1] Douglas B. West, *Introduction to Graph Theory*, Prentice Hall.
- [2] Ian Foster, *Designing and Building Parallel Programs*.
- [3] George Karypis and Vipin Kumar, “Fault-Tolerant Communication Topologies in Distributed Systems.”